

BLACKVILLE 3 W, SC, USA

WMO: 747840

Lat: 33.3550N

Lon: 81.3279W

Elev: 317

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.1	27.7	8.3	8.5	29.7	12.9	10.7	34.0	15.1	47.4	13.9	49.7	2.8	N/A	0.326

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	94.6	75.2	92.6	75.0	90.6	74.6	78.5	88.3	77.7	87.6	77.0	86.7	5.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.0	137.5	82.8	75.1	133.1	82.0	74.2	129.2	81.2	42.2	88.5	41.2	87.6	40.4	87.0	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.6	10.6	8.9	DB	17.8	97.6	3.7	3.3	15.1	99.9	13.0	101.8	10.9	103.7	8.3	106.0
			WB	16.3	80.5	4.4	1.0	13.1	81.1	10.5	81.7	8.1	82.2	4.9	82.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	46.1	49.5	56.1	63.7	71.8	78.3	80.3	79.4	74.9	65.3	54.5	49.3
	DBStd	14.07	10.08	9.01	8.56	7.00	5.67	3.73	3.16	3.45	4.60	7.29	8.05	8.98
	HDD50	526	200	111	42	3	0	0	0	0	0	2	42	126
	HDD65	2343	587	438	294	106	14	0	0	0	2	87	325	490
	CDD50	5702	79	97	229	414	677	849	940	910	748	475	178	106
	CDD65	2048	2	5	17	67	226	399	475	445	300	96	11	5
	CDH74	17905	7	29	165	682	2041	3684	4476	3793	2243	701	71	13
	CDH80	7258	0	0	17	151	749	1652	2064	1630	832	159	4	0
(14) Wind	WSAvg	3.8	4.4	4.6	4.7	4.3	3.9	3.4	3.1	2.7	3.2	3.3	3.6	4.0
(15) Precipitation	PrecAvg	47.6	4.5	3.9	4.5	3.1	3.6	5.4	4.7	4.8	3.5	2.9	2.4	3.4
	PrecMax	75.1	8.3	8.3	9.7	13.0	10.4	11.5	9.6	12.1	10.1	14.7	8.0	8.1
	PrecMin	33.2	0.7	0.7	1.0	0.4	0.7	0.7	1.2	0.9	0.1	0.0	0.7	0.8
	PrecStd	8.1	1.9	1.9	2.3	2.2	2.1	2.6	2.1	2.6	2.4	2.8	1.6	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	78.0	82.4	87.1	94.2	96.5	96.7	96.6	93.3	87.9	80.3	76.1
		MCWB	63.4	64.5	63.8	67.9	71.1	75.4	76.1	75.6	72.9	73.9	66.6	68.2
	2%	DB	69.9	73.1	78.5	84.2	90.2	93.7	94.4	93.1	90.1	84.2	75.8	71.7
		MCWB	60.8	61.8	63.3	66.2	70.4	74.9	76.1	76.0	73.5	70.4	63.6	64.6
	5%	DB	66.4	69.0	75.2	81.4	87.4	91.3	92.3	90.8	87.8	81.2	72.1	68.2
		MCWB	59.0	59.3	61.2	65.6	69.8	74.2	76.0	75.5	72.5	68.5	61.5	62.7
	10%	DB	62.6	65.0	71.4	78.1	84.4	88.9	90.0	88.5	85.3	77.9	68.9	64.7
		MCWB	56.6	57.7	59.8	64.0	69.1	74.2	75.9	75.3	72.5	66.7	61.2	60.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.1	67.5	67.9	72.5	74.8	78.5	80.1	79.8	77.8	76.4	69.5	70.0
		MCDB	70.5	74.0	76.5	81.6	85.0	89.5	91.1	88.8	86.1	82.9	74.0	73.6
	2%	WB	63.9	65.1	66.0	70.4	73.3	77.4	78.5	78.4	76.5	74.4	67.1	67.4
		MCDB	69.2	72.1	74.8	78.3	83.1	89.0	89.3	87.2	84.9	80.6	71.1	70.6
	5%	WB	60.9	63.0	64.2	68.9	72.3	76.5	77.8	77.5	75.3	72.6	65.3	64.2
		MCDB	64.8	69.2	72.2	76.7	81.9	87.8	88.4	86.4	83.7	78.0	69.8	67.2
	10%	WB	57.8	60.0	62.0	67.1	71.2	75.4	77.0	76.8	74.1	70.2	62.5	61.2
		MCDB	61.2	65.5	69.3	75.1	80.6	86.4	87.3	85.3	81.9	75.3	67.5	64.7
(35) Mean Daily Temperature Range	5% DB	MDBR	20.3	21.1	22.4	23.1	21.4	19.6	19.0	17.7	18.6	21.0	22.2	19.8
		MCDBR	22.9	24.4	25.8	26.5	25.5	23.0	21.9	21.2	22.3	22.9	24.6	21.0
		MCWBR	15.1	14.3	13.0	11.1	9.3	7.8	7.4	7.4	8.4	11.0	14.4	13.7
	5% WB	MCDBR	18.5	20.6	21.3	19.7	19.8	20.2	19.2	17.9	18.7	17.9	18.7	16.7
		MCWBR	14.3	14.3	12.1	10.4	8.0	7.9	7.5	7.3	7.9	10.3	13.8	13.5
(40) Clear-Sky Solar Irradiance	taub		0.324	0.332	0.367	0.403	0.470	0.508	0.550	0.526	0.437	0.400	0.347	0.320
	taud		2.487	2.463	2.393	2.285	2.142	2.082	1.986	2.042	2.280	2.345	2.460	2.535
	Ebn at Noon		283	292	288	281	262	251	240	244	263	264	271	278
	Edh at Noon		28	31	36	42	49	52	57	53	40	35	28	25
(44) All-Sky Solar Radiation	RadAvg		901	1100	1441	1818	1961	2010	1932	1759	1519	1283	987	786
	RadStd		46	110	108	127	142	140	135	91	129	150	74	72

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+410	+272
(47) Regional (53 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+340	+236

Nomenclature: See separate page